

Janadri Yalla Yashwanth

Software & DevOps Engineer | AWS · Docker · CI/CD · Linux · Python | Cloud & Infrastructure Automation

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SUMMARY

B.Tech CSE (AI & ML) 2025 graduate, CGPA 8.6, who builds, ships, and operates production systems on Linux — Docker & Docker Compose, GitHub Actions CI/CD, systemd services, AWS S3, and distributed task workers (Celery/Redis). Experienced across the full delivery cycle: development, containerization, cloud deployment, automation, and production troubleshooting (CUDA, DB migrations, GPU/CPU workers). Strong problem-solving, automation, and documentation. Immediate joiner.

TECHNICAL SKILLS

Cloud / Infrastructure: AWS (S3, /vsis3 streaming), Docker, Docker Compose, Linux (Ubuntu, Raspberry Pi OS), systemd

DevOps / Automation: GitHub Actions CI/CD, deployment automation, dependency packaging, shell & Python scripting

Languages: Python, Bash, C, SQL, Java

Backend / Distributed: FastAPI, Celery, Redis, PostgreSQL, REST APIs, async workers

Systems / Tooling: Linux administration, Windows, Git, monitoring, debugging & root-cause analysis

Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, networking basics

Tools: Git & GitHub, Claude Code, VS Code, Docker, GitHub Actions, AWS CLI, systemd, Linux

EXPERIENCE

Software Engineer Trainee — Marut Drones

Feb 2026 – Present

- Software engineer on the drone AI/automation team — shipped production computer-vision detection services, geospatial backend features, and autonomous-flight tooling; selected projects below.

Machine Learning Intern — Rooman Technologies

Jan 2025 – Mar 2025

- Built ML solutions for healthcare; developed and evaluated a sleep-disorder classification model on physiological and behavioral data, achieving 98% accuracy.

KEY PROJECTS — MARUT DRONES

Marut Survey Platform — Backend & AI Detection Services

FastAPI, Celery, Redis, PostgreSQL, PyTorch, SAM-2, S3/GDAL, Docker

- Built three production detection pipelines (stockpile, tree-crown, building) integrating three model families — SAM-2, DeepForest, and GroundingDINO/LangSAM — behind a FastAPI backend with async Celery/Redis job APIs (live progress, cancellation, owner-scoped RBAC).
- Derived per-object stockpile volumes (m3) and material classification from nDSM height integration, backed by a 7-factor confidence rubric and Union-Find cross-tile NMS dedup at 0.3 IoU.
- Streamed cloud-optimized GeoTIFFs block-by-block from S3 via GDAL /vsis3/ + rasterio (no full-raster load) and tuned S3 multipart transfers — 64 MB chunks, 25 concurrent threads, pooled connections — for multi-gigabyte orthomosaics.
- Packaged the PyTorch / SAM-2 / DeepForest / GroundingDINO stack into a Docker image shipped through GitHub Actions to systemd-managed GPU and CPU workers; Pydantic v2 schemas and Alembic migrations.

Flight Log Analyzer — Full-Stack Drone Diagnostics Platform

React, TypeScript, FastAPI, Python, Docker

- Built a full-stack web app that ingests raw drone flight logs and auto-generates professional PDF health/crash reports, supporting both ArduPilot (.bin) and JIYI K++ formats with automatic binary-vs-text detection via magic-byte sniffing.
- Engineered a from-scratch binary parser for the ArduPilot/DataFlash self-describing log format using Python struct, decoding 50+ message types and reconstructing real-world IST timestamps from GPS-week data.
- Implemented a phased diagnostic engine running 50+ automated checks (power, IMU/vibration, GPS, EKF, compass, RC, failsafes), classifying findings as PASS/WARN/FAIL and inferring probable crash cause.
- Developed automated PDF reporting with ReportLab + Matplotlib (flight-path maps, sensor charts, color-coded tables); built the React 19 + Vite + Tailwind front end with file upload and download handling.
- Containerized the full stack with Docker / Docker Compose; integrated a .NET converter via Mono + Xvfb inside Linux (~7,200 LOC, solo-built).

Autonomous Precision Landing on Moving Platform

ArduPilot, MAVLink, Raspberry Pi 5, AprilTag, OpenCV, Python

- Built an autonomous drone precision-landing system that lands on an AprilTag target on stationary and moving (ground-rover) platforms; achieved proven stationary-tag landings at ~5–10 cm accuracy.
- Wrote the Raspberry Pi 5 companion-computer controller (pupil-apriltags + picamera2 + OpenCV) streaming LANDING_TARGET and velocity setpoints to a CubeOrange (ArduCopter) over MAVLink (pymavlink) at 20 Hz.
- Designed a three-mode state machine — visual P-control with rover-velocity feed-forward, GPS-NAV haversine approach, and hover fallback — with pilot RC override, velocity caps, and touchdown auto-disarm.
- Implemented multi-vehicle MAVLink (drone + rover SYSIDs), camera calibration, systemd auto-start, and a BENCH safety mode for props-off bench testing.

EDUCATION

Sri Venkateswara College of Engineering, Tirupati

Dec 2021 – May 2025

B.Tech in Computer Science (Artificial Intelligence & Machine Learning) — CGPA: 8.6

Narayana Junior College, Hyderabad

Jun 2019 – Mar 2021

Intermediate (MPC) — 96%

CERTIFICATIONS

- CS50's Introduction to Programming with Python — Harvard University Online
- Databases: Topics in SQL — Stanford Online